

Cognitive Opportunistic LEO-PNT Receiver System - DhruvXPNT

Detailed Technical Description and Design Approach

(PSID 25217 – Ministry of Defence / HQ IDS)

1. Background and Motivation

Position, Navigation, and Timing (PNT) services provided by conventional Global Navigation Satellite Systems (GNSS) are a critical enabler for defence operations, supporting navigation, command and control, secure communications, precision engagement, and synchronized mission execution. Despite their widespread adoption, GNSS systems are inherently vulnerable to a range of threats and limitations. These include deliberate jamming and spoofing, signal attenuation due to terrain or built-up environments, multipath interference, atmospheric effects, and degradation caused by poor satellite geometry or satellite anomalies. In modern conflict scenarios, GNSS denial or degradation is no longer a theoretical risk but a practical reality.

As a result, there is a growing operational requirement for alternative or complementary PNT solutions that can operate in GNSS-denied or GNSS-degraded environments. Emerging research and experimental systems indicate that Low Earth Orbit (LEO) satellite constellations can play a significant role in addressing this gap. The increasing number of LEO satellites across multiple constellations has led to a dense and diverse RF environment, which can be exploited to derive navigation observables without relying on traditional navigation signals.

2. Concept of Signals of Opportunity for LEO-PNT

Signals of Opportunity (SoO) refer to radio frequency transmissions that are not specifically designed for navigation purposes but can be exploited to extract PNT-related measurements. In the context of LEO satellites, these signals typically originate from communication, data relay, IoT, or Earth observation platforms. Unlike GNSS signals, these transmissions often have unknown or proprietary signal structures and are supported by non-navigation-grade clocks and orbits.

However, LEO satellites exhibit very high relative velocities with respect to ground receivers, typically on the order of 7–8 km/s. This results in strong and rapidly varying Doppler frequency shifts, which can be accurately measured at the receiver. By tracking these Doppler signatures over time and correlating them with predicted satellite motion, it is possible to estimate the receiver's position, velocity, and clock parameters without decoding the underlying data payload. This physical-layer exploitation approach forms the foundation of opportunistic LEO-based navigation.

3. Proposed System Overview

The proposed system is a Cognitive Opportunistic LEO-PNT Receiver designed to provide GNSS-independent or GNSS-resilient PNT capabilities. The receiver adopts a software-defined and frequency-agnostic architecture, allowing it to adapt dynamically to different

LEO signal sources and operating environments. The system is intended to operate continuously for extended durations, with a target of 24-hour operation, and to be realizable in a handheld or low size, weight, and power (SWaP) form factor.

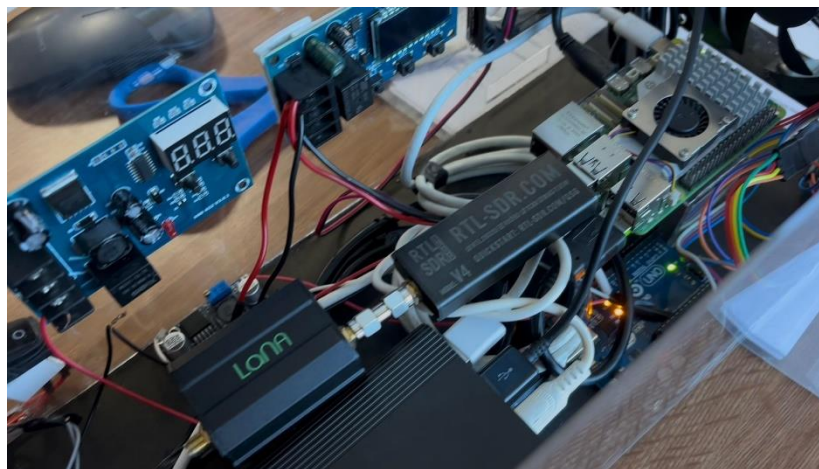
The receiver does not require cooperation from satellite operators and does not depend on navigation message decoding. Instead, it focuses on extracting Doppler-based and carrier-frequency-based observables from received RF signals. Cognitive capabilities are incorporated to enable adaptive signal selection, quality assessment, and robust operation in the presence of interference and uncertainty.

4. System Architecture

4.1 RF Front-End

The RF front-end is responsible for capturing LEO satellite signals across one or more frequency bands. It consists of antennas optimized for LEO reception, a low noise amplifier (LNA) to improve signal-to-noise ratio, and a software-defined radio (SDR) capable of wideband raw IQ sampling. Active antennas may be supported using bias-tee power injection where required.

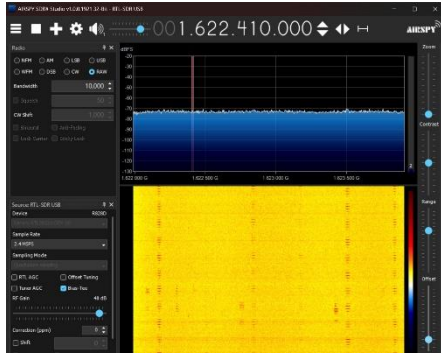
Key design considerations include frequency stability, dynamic range, and portability. While navigation-grade oscillators are not mandatory, improved receiver frequency stability (e.g., TCXO or OCXO) significantly enhances Doppler estimation accuracy and long-term stability.



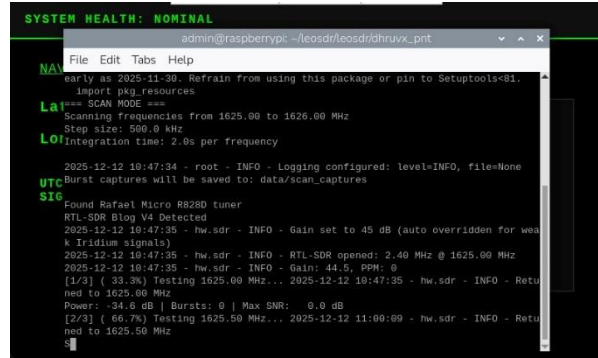
4.2 Signal Acquisition and Preprocessing

The receiver performs continuous or scheduled spectrum sensing to identify candidate LEO signals. Since the signal structures are generally unknown, acquisition is based on energy detection, spectral characteristics, and temporal behavior rather than protocol-specific synchronization.

Time–frequency analysis techniques, such as the Short-Time Fourier Transform (STFT), are employed to isolate narrowband or carrier-dominant components suitable for Doppler tracking. Burst detection and signal segmentation are used to identify valid observation intervals while rejecting noise-dominated or highly corrupted segments.



Airspy SDR software for testing antenna reception

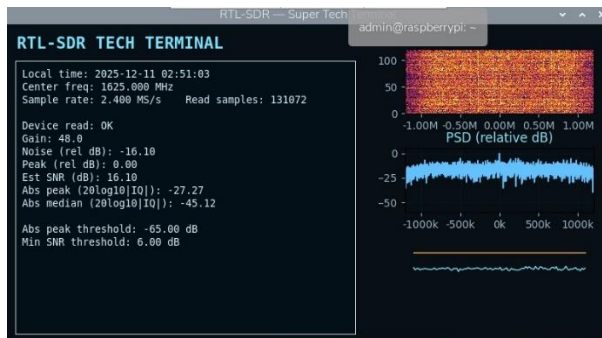


DhruvXPNT system signal acquisition and preprocessing

5. Doppler Estimation and Measurement Processing

Doppler estimation forms the core measurement process of the opportunistic LEO-PNT system. For each detected signal, the receiver tracks the carrier frequency variation over time and estimates the instantaneous Doppler shift. The Doppler rate and higher-order curvature are also analyzed to assess measurement consistency and quality.

Unlike GNSS, where absolute frequency accuracy is critical, the proposed approach emphasizes relative Doppler variation. This significantly reduces sensitivity to satellite clock instability and unknown frequency offsets. Measurements exhibiting non-linear behavior, excessive noise, or interference-induced distortions are flagged and rejected by the cognitive processing layer.



6. Orbit Modeling and Expected Doppler Computation

Satellite orbit information is obtained from publicly available Two-Line Element (TLE) data. Orbit propagation is performed using established models such as SGP4 to compute satellite position and velocity over time. Using an assumed receiver state, the expected Doppler shift is calculated based on relative line-of-sight velocity between the satellite and receiver.

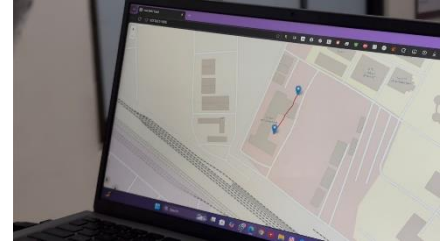
Due to inherent inaccuracies in TLE data and orbit propagation models, orbit-induced errors are explicitly accounted for in the navigation estimation process rather than assumed to be negligible.

7. Navigation Estimation Framework

The navigation solution is obtained by minimizing the residuals between measured and predicted Doppler values across multiple satellites and observation intervals. Estimation is

performed using a least-squares or Extended Kalman Filter (EKF) framework, which allows joint estimation of receiver position, velocity, and clock parameters.

Typical state variables include receiver position, receiver velocity, receiver clock bias, and receiver clock drift. The filter structure allows incremental refinement as new Doppler measurements become available and supports continuous navigation over extended durations.



8. Cognitive and Adaptive Capabilities

Cognitive functionality is integrated throughout the receiver to enhance robustness and accuracy. The system dynamically evaluates signal quality based on metrics such as carrier-to-noise ratio, Doppler linearity, observation duration, and satellite geometry. Based on these assessments, the receiver selects the most informative signals and satellites for navigation estimation.

This adaptive approach enables the receiver to operate effectively in complex RF environments, automatically avoiding heavily interfered bands and prioritizing signals that provide the highest navigation observability.

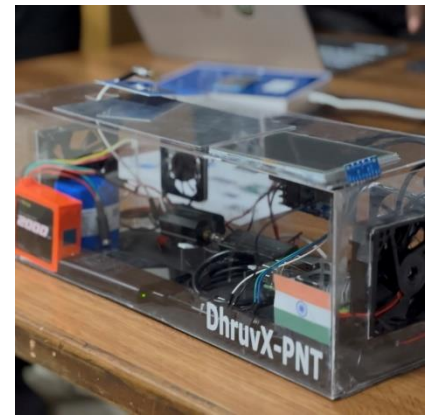
9. Error Sources and Mitigation

The opportunistic LEO-PNT approach is subject to several error sources, which are addressed through modeling and estimation strategies.

Error Source	Description	Mitigation Approach
Satellite orbit error	Inaccuracies in TLE-based propagation	Multi-pass averaging, residual filtering
Satellite clock instability	Non-navigation-grade clocks	Relative Doppler processing
Receiver oscillator drift	Frequency drift over time	Clock bias and drift states in EKF
Atmospheric effects	Minor Doppler distortion	Elevation masking, modeling
Multipath and interference	Measurement corruption	Quality metrics and signal rejection

10. Sensor Integration and Continuity

To improve continuity and robustness, the receiver integrates auxiliary navigation sensors such as inertial measurement units (IMU) and barometric sensors. These sensors provide short-term motion propagation and vertical observability, allowing navigation continuity during brief signal outages or poor satellite visibility. Sensor fusion is naturally incorporated within the EKF framework.



11. Validation and Performance Assessment

System validation is carried out through a combination of static and dynamic field experiments. Performance is evaluated by comparing opportunistic LEO-PNT solutions against reference GNSS solutions when available. Key metrics include position error, convergence time, Doppler residual statistics, and stability over long observation periods. Additional testing under partial or simulated GNSS denial conditions is used to assess resilience and graceful degradation.

12. End Objective

The ultimate objective is to realize a deployable, low-SWaP Cognitive Opportunistic LEO-PNT receiver capable of providing reliable PNT services in GNSS-challenged environments. The system is intended to function as a standalone navigation aid or as a complementary layer alongside GNSS and inertial navigation systems, enhancing overall resilience for defence applications.